Behavioral and Autonomic Responses to Peer Separation in Pigtail Macaque Monkey Infants

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Brief maternal separations of young nonhuman primates have been used extensively to study the behavior and physiology of attachment, loss, and bereavement. The physiological responses to the loss of alternative attachment figures, such as peers, is less well documented in nonhuman primates. This study examined both autonomic and behavioral responses of peer-reared pigtail macaque infants to separation. Eight infants were removed from their mothers at birth and reared in four peer pairs. At 6 months of age, each monkey was implanted with a multichannel biotelemetry device which transmitted heartrate, body temperature, EEG, EMG, and EOG. Blood was collected twice weekly for immunological assessment. Behavioral and physiological data, including sleep, were collected for 1 week of baseline, 2 weeks of separation, and 1 week of reunion. Behavioral and physiological results indicated agitation but not depression following separation from their peer attachment figures. We found reduced mitogenic responses to pokeweed consequent to peer separation, suggestive of altered B-cell function. REM variables were the only sleep measures affected by the separation, and were suggestive of agitation but not depression.

Introduction

The social separation paradigm in nonhuman primates has been an important experimental approach to the study of the mother-infant relationship, and social attachment generally (Mineka & Suomi, 1978). Separation has also been used as an animal model of human loss, depression, and bereavement (Kaufman & Rosenblum, 1967a,b; McKinney & Bunney, 1967; Reite, Short, Seiler, & Pauley, 1981). The types of relationships that have been studied using the separation

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paradigm include mother-infant, peer-peer, nuclear family, and infant-surrogate (Hennessy, 1986; Kaufman & Rosenblum, 1967a,b; Mineka & Suomi, 1978; Reite et al., 1981; Reite, Short, & Seiler, 1978; Seay, Hanson, & Harlow, 1962; Suomi, Eisele, Grady, & Harlow, 1975).

The results of mother-infant separation studies in nonhuman primates have frequently yielded results similiar to the responses observed in human infants and children, which has been termed the "protest-despair" reaction (Bowlby, 1960). This response involves an initial agitation (or "protest") phase characterized, in primates, by increased locomotion and distress vocalizations, followed by a second "despair" or depression phase, characterized by a decrease in activity and social play, and facial expressions and postures suggesting depressed affect (Bowlby, 1960; Hinde, Spencer-Booth, & Bruce, 1966; Kaufman & Rosenblum, 1967a,b; Seay et al., 1962). Studies of the infant's physiological response to mother-infant separation experiences have reported disturbances in the regulation of heart rate, body temperature, and sleep patterns in pigtail infants (Reite et al., 1981), lymphocyte transformation in bonnet (*M. radiata*) infants (Laudenslager, Reite, & Harbeck, 1982), specific antibody production (Coe, Rosenberg, & Levine, 1988; Laudenslager, Reite, & Held, 1986), and endocrinological measures in squirrel monkeys (Coe, Glass, Weiner, & Levine, 1983; Hennessy, 1986) and rhesus macaques (Gunnar, Gonzalez, Goodlin, & Levine, 1983).

A number of studies have used peer separation to model depression, citing potential advantages such as improved accessibility of the subject, ability to perform repeated separations, and absence of a developmentally changing mother-infant relationship as a possible confound (Suomi, Harlow, & Domek, 1970; Lewis, McKinney, Young, & Kraemer, 1976; Mineka & Suomi, 1978). Results by researchers examining peer separation responses are contradictory, however. Some researchers working with rhesus macaques (Mineka, Suomi, & DeLizio, 1981; Suomi et al., 1970) report a two stage response in peer separations similar to what has typically been found in mother infant separations. Others (Anderson, 1983; Erwin, Mobaldi, & Mitchel, 1971; Maxim, 1980; Mineka, Gunnar, & Champoux, 1986; Reite, Harbeck, & Hoffman, 1981) report an agitation phase, but no evidence of behavioral depression in response to peer separation. This may be attributable, in part, to species or methodological differences in rearing or separation procedures.

There are fewer studies of the physiological correlates of peer separation in monkeys. Brief (30 min) peer separations in young squirrel monkeys resulted in elevations of plasma cortisol similar to that found for the "protest" response of maternally separated infants (Hennessy, Mendoza, & Kaplan, 1982). A pilot study in a pair of pigtail infants observed a decrease in the lymphocyte response to mitogen stimulation during a 14-day separation from a peer. Responses returned to baseline levels on reunion of the peers (Reite et al., 1981).

In this article we examine a comprehensive set of physiological and behavioral variables during peer separations in infant pigtail macaques.

Methods

Subjects

Our subjects were eight pigtail macaques born in the Department of Psychiatry Primate Laboratory (Table 1). The infants were separated from their mothers

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TABLE 1. Subject Descriptions.

Animal	Peer Pair	Sex	Date of Birth	Age at Initial Separation from Mother (days)	Age at Weaning (days)	Age at Implant (days)	Age at Separation (days)
23.3.2	1	*F*	147–84	4	173	207	230
53.6.3	1	*M*	148–84	3	172	150	229
17.2.4	2	*M*	5–85	2	158	174	198
63.2.4	2	*F*	20–85	2	149	153	183
32.6	3	*M*	48–85	3	129	191	218
63.4.2	3	*M*	49–85	2	128	190	217
30.3	4	*F*	91–85	14	175	191	218
50.7.2*	4	*F*	91–85	15	175	191	218
Mean				5	157	182	214
SD				6	20	19	16

*This animal's transmitter was nonfunctional.

at a mean age of 4 days, and were reared together in pairs with a peer who differed in age by no more than 15 days. Each peer pair lived in a 1 m^2 cage for the first 2 months of life. Fluorescent lights (700 lux) were off between 2000 and 0700 hours. Infants were initially fed a milk-based formula (Enfamil), and were weaned to a fruit and monkey chow diet by 155 days. They were fed at 0900 hours daily, with water available *ad libitum*. At about 175 days of age each pair was moved into a set of two adjoining pens measuring 1.5 × 2.5 × 3 m each, connected by a small 1-m long tunnel, which normally provided the animals free access to both pens. The animals used both pens freely during the baseline period. Separation was effected by closing the tunnel between pens, leaving each animal alone in one of the (familiar) adjoining pens. The animals, while separated, could hear, but could not see, smell (as the two pens were on separate ventilation systems), or touch each other.

Behavior

Behavioral observations were recorded by trained observers directly online using a PDP 11/23 computer (Walker & Reite, 1978). The behavioral data collection system allowed tabulation of durations and frequencies of social and nonsocial behaviors derived from the behavioral taxonomy developed by Kaufman and Rosenblum (1966). Behaviors discussed in this article are listed in Table 2.

Behavioral observations were made from behind one-way mirrors during six 5-minute daily sessions, three times each between 0800 and 1200, and 1300 and 1700.

Physiology

At a mean age of 180 days, the infants were sedated with Ketamine, anesthetized with IV sodium pentobarbital 15 mg/kg, and surgically implanted with multichannel biotelemetry systems that transmitted seven channels of physiological data; EOG, EMG, EKG, body temperature, and three channels of

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TABLE 2. Behavior Definitions.

Behavior	Definition
Activity = the sum of:	
Movement	Any activity
Locomotion	Locomotor movement in any direction of at least 6 inches involving limb movement
Activity level	Number of times infant crossed imaginary border, dividing pen into 9 equal cubes
Minus the sum of:	
Quiet Rest	At rest, eyes closed
Active rest	Some movement, eyes open or closed
Environmental Exploration = the sum of:	
Bedding Explore	Visual or manual inspection of bedding materials
Object Explore	Visual or manual inspection of anyother environmental component
Oral Object	Oral manipulation of environment
Inattentive Object Explore	Exploration of environmental object while visual orientation is away from the object
Active Search	Explore environment visually with movement
Vocalization = the sum of:	
Coo	Long, tonal vocalization
Squeal	Gecker vocalization
Self-Directed Behaviors	
Self-orality	Self mouthing or self-licking
Self-groom	Grooming directed to self
Scratch	Common usage
Self-play	Playing alone
Social Behaviors:	
Social Play	Playing with other infants
Cradle	Mutual clinging
Follow	Common usage
Mount	Animal places hands on hip of other and may or may not include foot-clasping ankles of mounted animal
Groom	Manual/oral manipulation of other animal's fur, skin, etc.
Lipsmack	Rapid opening and closing of lips

EEG (Pauley & Reite, 1981). Both members of each peer pair were implanted the same day and recovered together. The animals wore leather vests following surgery until sutures were removed 7–10 days later. Following recovery, the implanted telemetry units were activated and collection of physiological and behavioral data began. Animal 23.3.2's transmitter was nonfunctional and was subsequently replaced; 23.3.2's peer (53.6.3) was anesthetized during the second implant procedure. Animal 50.7.2's transmitter failed following the first day of separation, leaving seven animals in the sample for telemetry data. Its pair partner (30.3) was observed without disturbance for turning the transmitter on/off or blood sampling.

While the units were on, physiological data were transmitted 24 hr/day. An on-line computer system (Reite & Short, 1983) sampled HR and BT every minute, and an observer took HR and BT readings at 15-minute intervals from a Grass

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Model 78 polygraph. Physiology variables included measures of nocturnal sleep, estimates of HR and BT circadian rhythms, and measures of mean day and mean night HR and BT. Mean day HR and BT were computed by averaging all samples obtained from 1000 to 1600 hours; mean night HR and BT were the mean of all samples obtained between 2200 and 0400 hours.

As a means of assessing the circadian component of HR and BT, 24-hour cosine curves were fit (least squares technique) to the computer-collected HR and BT samples from 0900 one day to 0900 the next (Halberg, Tong, & Johnson, 1967; Reite & Short, 1981). Cosine variables obtained in this manner included daily measures of the level, amplitude, and phase of the best fit cosine. The level approximates the mean of the 24-h data. The amplitude represents the maximum deviation of the best fit cosine above or below the level (or approximately half the maximum day-night difference). The phase, or "acrophase," represents the time in hours (from 0900) to the peak of the best fit cosine.

On selected nights, during the lights-off period, physiological data were recorded continuously on the polygraph for scoring of nocturnal sleep. The resulting paper sleep record was hand scored to provide measures of the sleep variables previously described for pigtail infants (see Reite & Short, 1978, for details). Sleep variables scored (in minutes) included sleep latency, REM latency, each sleep stage (Drowsy, Stage 2, Stage 3 and 4, REM), inter-REM interval, mean REM period length, and total sleep time, as well as number of arousals and REM periods.

Immunology

Periodic blood samples were rapidly obtained by venipuncture from all animals under light restraint in small transport cages. The *in vitro* proliferative responses of the lymphocytes to B cell (pokeweed mitogen [PWM]) and T cell (phytohemagglutinin [PHA] and concanavalin A [ConA]) mitogens were studied. The lymphocytes were removed by standard *Ficoll-Paque* sedimentation and resuspended in supplemented RPMI-1640 media at a concentration of 1×10^6 cells/mL. Lymphocytes (0.1 mL) were incubated in triplicate in wells of a microtiter plate for 72 hr at 37°C in a 5% humidified CO_2 incubator with varying concentrations for each mitogen. These represented optimal and suboptimal concentrations. At the end of 72 hr, 10μCi of ^{3}H-thymidine was added to each well, and 4 hr later cells were harvested onto glass wool filter paper. The filter discs were placed in scintillation fluid and counted on a liquid scintillation counter. Results were expressed as counts per minute of experimental mitogen-stimulated cultures (E) minus counts per minute for unstimulated cultures (C) $\times$ 10^{-3} $[(E - C) \times 10^{-3}]$. Additionally, total white cell counts and differential cell counts were determined for all subjects.

Experimental Protocol

Beginning at approximately 7 months of age, 5 days of baseline behavioral and physiological measures were collected. Following baseline, members of the pair were separated for 14 days. During 10 of these days, behavioral and physiological data were collected. On the 15th day, peers were reunited (by opening the tunnel)

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and 5 more days of behavior and physiology were recorded. These conditions will be referred to as follows in the remainder of the article: B (baseline), SD1 (separation day 1), S1 (remainder of first week of separation), S2 (second week of separation), S (all of separation except SD1), RD1 (reunion day 1), R (remainder of reunion). All physiological variables were not available from all animals on all days for technical reasons.

All night sleep records were recorded simultaneously from both members of a pair for 4 nights during B, SD1, 4 separation nights (two randomly selected from each week of S), RD1, and the 2 nights following reunion. All infants except Pair 4 had a 2–3 mL blood sample taken when the transmitter unit was turned on at 0800 on Monday of each week of data collection. Transmitters were switched off each Saturday after 0900 hours. Peer pair 4 had no blood samples taken, and 30.3's transmitter unit remained on for the entire experiment.

Analysis

Behavior

To reduce the number of variables analyzed and minimize intercorrelations between behaviors, summary categories were created by combining mutually exclusive behavior categories which were highly correlated with one another (Table 2). Not all the categories in Activity were mutually exclusive: activity was counted regardless of the animal's behavior, although was only possible when the animal was in locomotion or play vigorous enough to carry it further than 60 cm. Hence, there was only limited overlap between even these categories and their sum was informative regarding activity.

For each experimental group, repeated measures Analyses of Variance (ANOVAs) with experimental condition as the repeated measures factor, were conducted for these summary behaviors, and Newman–Keuls post hoc tests were used to identify significant differences for significant ANOVAs.

Heart Rate and Body Temperature

Measures of HR and BT circadian values, and mean day and night values were averaged for B, S1, S2, and R. Repeated measures Analyses of Variance (ANOVAs) with period (B, SD1, S1, S2, RD1, R) as the repeated measures factor were performed for the HR and BT variables.

Sleep

Nocturnal sleep was scored from the paper record using criteria developed previously for monkey infants (Reite, Stynes, Vaughn, Pauley & Short, 1976). All sleep records could be scored for at least 95% of the night. Sleep variables were averaged for B, S, and R, with SD1 and RD1 treated separately. Repeated measures ANOVAs, with condition (B, SD1, S, RD1, R) as the repeated measures factor were performed for each sleep variable.

Preliminary analyses included peer pair as a factor in the ANOVA to examine

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the impact of different pairs on the results. These analyses found that membership in a peer pair did not affect the outcome (main effects or interactions). Therefore, this factor was dropped from the analysis and ANOVAs were conducted with individual subject's data.

Results

The peer-reared monkeys in this study exhibited both behavioral and physiological evidences of agitation in response to separation from their peer partner. There was no evidence of depression following this phase, and by the second week of separation there was some evidence of behavioral recovery.

Activity Measures

The infants were more active during separation (particularly on the first day of separation) than during baseline and reunion conditions, indicating the presence of agitation in response to separation [$F = 8.69$, 5/35 df, $p < .001$; Fig. 1(a)].

Environmental exploration also increased dramatically during the separation phase of the experiment. Both duration and frequency showed increases across the separation, such that the highest levels occur in S2 [duration: $F = 4.79$, 5/35 df, $p < .01$; frequency: $F = 17.08$, 5/35 df, $p < .001$; Fig. 1(b)].

Eating and drinking behaviors both declined on SD1 and returned to baseline levels during reunion. Eating also increased during S1 and S2 over baseline levels. [Drink: $F = 3.88$, 5/35 df, $p < .01$; Eat: $F = 8.03$, 5/35 df, $p < .01$; Fig. 1(c)].

Self-Directed Behaviors

Self-directed behaviors, indictive of agitation increased during the first week of separation (Self Groom $F = 5.24$, 5/35 df, $p < .01$; Scratch $F = 5.37$, 5/35 df, $p < .001$). Self-orality remained depressed throughout the whole separation (Phase $F = 3.09$, 5/35 df, $p < .05$), while Self-groom and Scratch returned to baseline levels during S2, indicating some adaptation to and recovery from the separation [Fig. 2(a)].

Self-play declined during the first week of separation, but increased to levels significantly higher than those during baseline during S2 [$F = 7.28$, 5/35 df, $p < .001$; Fig. 2(b)]. The levels of Self-play seen in S2 are near the levels of Social Play seen during baseline, suggesting that Self-play was replacing this behavior, as the animals began to recover from the separation, and return to baseline behaviors.

Social Behaviors

Vocalizations were close to zero during the baseline and increased dramatically on the first day of separation (to 176 freq/1000 sec), and then returned to near zero for the remainder of S1 and S2 ($F = 5.24$, 5/35 df, $p < .01$).

Several social behaviors decreased on RD1 then returned to baseline levels, including Mutual Cling, Follow, Mount, Groom, and Touch. Lipsmack, in contrast, increased on RD1 (Table 3).

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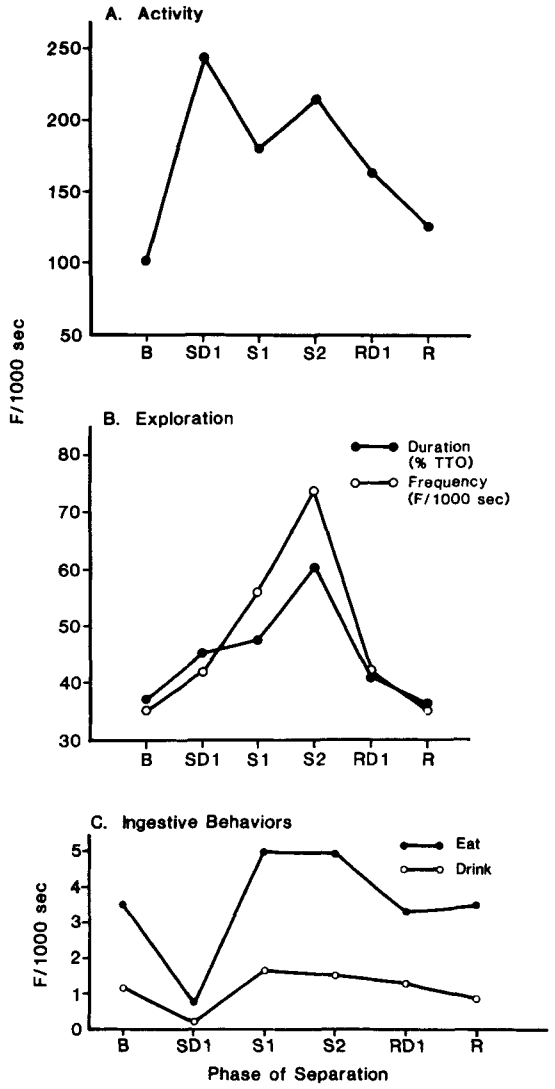


Fig. 1. Changes in the frequency of (a) Activity, (b) Environmental exploration, and (c) ingestive behaviors.

TABLE 3. Changes in Social Behaviors.

Behavior	Baseline	Reunion Day 1	Reunion Week	*F*-Ratio*	*p*-value
Cling	11.34	4.90	15.50	6.16	.012
Follow	0.35	0.00	0.22	3.73	.050
Mount	3.59	1.02	3.61	13.95	.0005
Touch	31.67	19.46	37.80	4.30	.035
Groom	0.14	0.05	0.28	4.23	.0366
Lipsmack	0.07	0.40	0.05	4.11	.0395

*2/14 degrees of freedom for all *F*-ratios.

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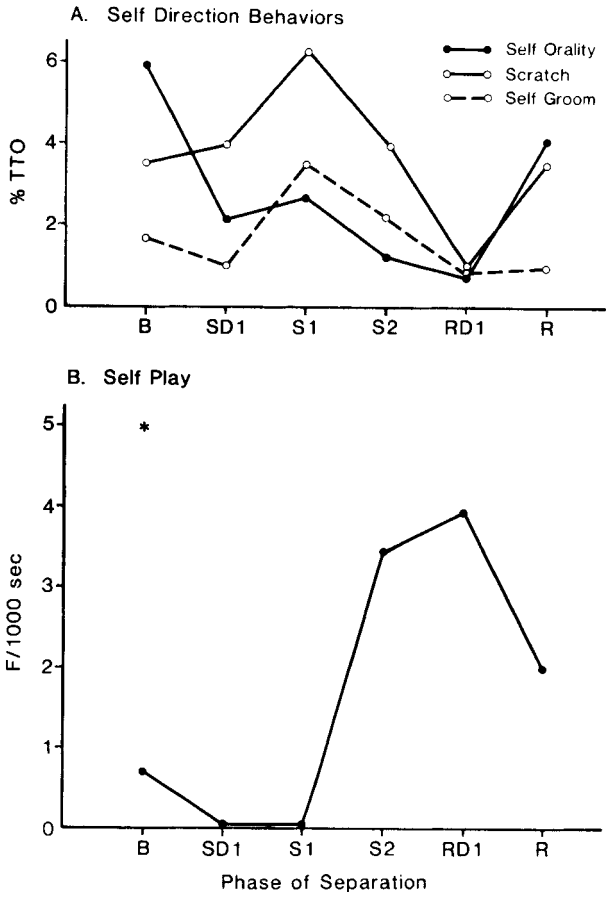


Fig. 2. Changes in (a) self directed behaviors, and (b) self play (Asterisks in (b) indicate baseline levels of social play).

Heart Rate and Body Temperature

With regard to HR and BT circadian rhythms, HR level was essentially unchanged throughout the experiment. HR amplitude was increased on SD1 and RD1 ($F = 9.62$, df = 5,20; $p < .001$) [Fig. 3(a)]. Additional analyses indicated that this increase was due to an increase in mean day HR on SD1 and RD1 ($F = 28.22$; df = 5,30; $p < .001$, followed by Newman–Keuls tests). Unlike HR, BT amplitude did not change significantly. Additional analyses indicated that BT mean day values were increased over baseline for all experimental conditions for both daytime values ($F = 9.77$; df = 5,30; $p < .001$, and night time values ($F = 3.91$; df = 5,30; $p < .01$). BT phase was delayed for about an hour on SD1, and then returned to baseline values ($F = 6.01$; df = 5,20; $p < .01$) [Fig. 3(b)].

Animal 30.3, which was not handled for either blood sampling or battery activation during the experiment, showed physiological changes similar to the other animals in this study, suggesting that subject handling probably contributed little to the continued agitation response.

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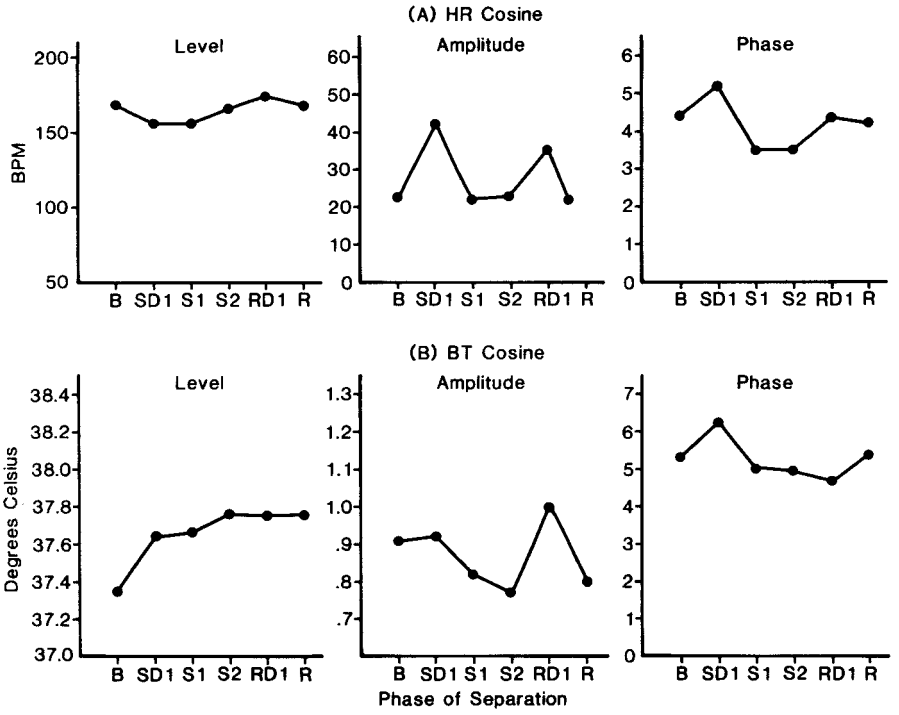


Fig. 3. Changes in circadian measures of (a) heartrate and (b) body temperature.

Sleep

Statistically significant changes were noted for sleep latency, and for four REM variables, including REM, #REMP, $\bar{x}$REML, and IRI (Table 4). Sleep latency was not effected by separation, but rather by reunion, showing a decrease on RD1, and an increase for the remainder of reunion ($F = 3.49$; df = 4,20; $p < .05$). The REM related changes on the other hand appeared to reflect both separation and reunion related experiences, with REM ($F = 7.78$; df = 4,20; $p < .001$), #REMP ($F = 4.01$; df = 4,20; $p < .05$), and $\bar{x}$REML ($F = 3.76$; df = 4,20; $p < .05$) decreasing with separation, and decreasing yet further with reunion. IRI was effected predominantly by an increase the first reunion night, when it was significantly different from both S1 and R ($F = 3.27$; df = 4,20; $p < .05$).

Immunology

Small changes were found for some immunological variables (i.e., B cell but not T cell mitogen responses [pokeweed mitogen (1.0 μg/well) $F = 8.48$; df = 2,14; $p < .01$; pokeweed mitogen (3.0 μg/well) $F = 5.98$; df = 2,14; $p < .05$]) (Fig. 4). The decrease in lymphocyte proliferation in response to PWM was noted during the second week of separation. Percentages of lymphocytes ($F = 4.33$; df = 2,14; $p < .05$) and segmented neutrophils ($F = 4.29$; df = 2,14; $p < .05$) indicated a change from the first baseline sample to subsequent samples such that the

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TABLE 4. Means of Sleep Variables for each Experimental Phase.

Variable	Baseline	First Night of Separation	Other Nights of Separation	First Night of Reunion	Other Nights of Reunion
Awake	135 ± 55	118 ± 74	129 ± 60	179 ± 56	123 ± 43
Arousals	66 ± 29	52 ± 36	53 ± 26	52 ± 25	53 ± 20
Sleep Latency	24 ± 14	22 ± 4	27 ± 11	16 ± 4	35 ± 11
TST	490 ± 59	514 ± 67	483 ± 59	436 ± 59	481 ± 41
Drowsy	10 ± 12	8 ± 9	7 ± 9	14 ± 31	8 ± 12
Stage 2	287 ± 47	295 ± 92	290 ± 73	251 ± 63	300 ± 40
Stage 3 & 4	144 ± 21	169 + 17	149 ± 17	162 ± 44	136 ± 34
REM*	59 ± 29	40 ± 30	39 ± 19	27 ± 11	40 ± 14
REM Latency	111 ± 29	187 ± 79	143 ± 38	145 ± 44	119 ± 61
#REMP*	7 + 1	7 ± 1	6 ± 1	5 ± 2	6 ± 1
XREML*	9 ± 2	7 ± 4	7 ± 2	6 ± 3	7 ± 3
IRI*	76 ± 5	69 ± 18	81 ± 18	107 ± 22	72 ± 34
REM EFF	79 ± 7	85 ± 7	80 ± 9	88 ± 15	83 ± 6
Sleep EFF	81 ± 9	83 ± 11	80 ± 10	76 ± 12	81 + 8

*ANOVA significant at $p < .05$.

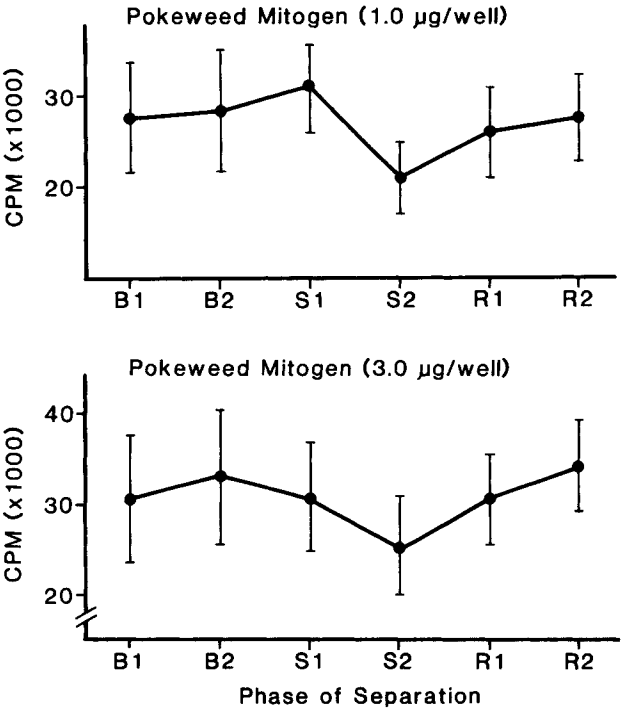


Fig. 4. Changes in the lymphocyte proliferative response to Pokeweed mitogen at concentrations of (a) 1.0 µg/well and (b) 3.0 µg/well.

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proportion of neutrophils was highest and lymphocytes lowest on that first sample.

Discussion

Peer separation in these pigtail monkeys did not result in the agitation-despair response typically seen in similar aged mother-reared monkeys and in humans (e.g., Bowlby, 1973; Seay et al., 1962; Reite & Capitanio, 1985; Reite et al., 1981). Instead, there was evidence only of physiological and behavioral agitation, which was sustained over the course of the separation. There was some evidence of behavioral recovery by the second week of separation, with a notable absence of a despair phase.

HR and BT changes were consonant with a persistent agitation-like reaction similar to the agitation seen in mother-separated infants (Reite et al., 1981). The subsequent decreases in both HR and BT most often found in maternally separated infants during the "depressive" response were not observed.

The sleep changes observed in these peer-separated pigtail infants seemed to be related more to the disruption associated with the return of the peer than the separation. Nocturnal sleep latencies were shorter on the first day of reunion. REM variables were also effected more by reunion than by separation, and the profound separation-related REM decreases seen in mother separated infants were not as apparent. These results are consonant with previously reported normative baseline nocturnal sleep observations, suggesting that the peer relationship per se is a significant disturbing factor in nocturnal sleep (Kaemingk & Reite, 1987).

Changes in immunological measures were small, and would be difficult to interpret as being of clinical relevance on an individual basis. The changes in proportions of neutrophils and lymphocytes from the first to subsequent samples probably represents a stress response to the first bleeding experience and subsequent adaptation to the procedure. The results of the mitogen stimulation studies suggests a modulation of the B cell population but not the T cells. Our prior study (Reite et al., 1981) observed in a smaller sample a change in response to T cell mitogens (PHA and ConA) associated with peer separation, whereas in the present study only the mitogenic response to the B cell mitogen (pokeweed) was significantly affected by the separation procedure. The variability in the mitogen response has been recently emphasized by Maier and Laudenslager (in press) and this apparent discrepancy may be reflective of the problems inherent in this assay. It is important to note, however, that the apparent depressed B cell response is consistent with the reduced antibody response, a B cell-mediated response, previously noted in association with mother-infant separation (Coe et al., in press; Laudenslager, Reite, & Held, 1986).

In summary, peer separation appears to produce a different set of physiological and sleep pattern changes than does maternal separation in pigtail macaque infants. Such findings suggest that the maternal and peer separation paradigms may differ in important ways. Two possible explanations need to be considered. First, the experimental protocol itself might have produced these differences. While the separated animals remained in familiar environments, they were (with

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the exception of animal 30.3) handled briefly twice weekly during the experiment to turn transmitters on and off and to obtain a blood sample. They had been habituated to such handling prior to the experiments, and therefore, this handling seems unlikely to account for the lack of depression. It is possible, however, that the arousal associated with handling may, despite the habituation, have contributed to an increase in sympathetic activity, promoting a state of agitation. The fact that unmanipulated animals (50.7.2 for behavior, and 30.3 for both behavior and physiology measures) also showed similar prolonged agitation, however, seems to contradict such a conclusion. The familiar environment of separation is not of itself likely to be the origin of the difference: the comparison data of mother-infant separations conducted in this lab were also conducted in familiar home-cage environments. This paradigm is used to specifically avoid confounding separation from the attachment figure per se with responses to a novel, and perhaps fear-provoking, environment.

Secondly, it may be that the attachment relationship in peer-reared infants is fundamentally different from mother-reared infants, and that this difference is reflected in a unique response to separation from the attachment object. Several other studies have also found fewer physiological effects of separation from nonmaternal attachment figures, including cortisol, HR, BT, and sleep measures in peer- and surrogate-reared infants (Hennessy, Mendoza, & Kaplan, 1982; Hennessy & Kaplan, 1982; Reite, Short, & Seiler, 1978). The problem remains, however, to explain prior peer separation studies which report despair (e.g., Mineka et al., 1981; Suomi et al., 1970). These latter separation studies are quite different, in that they entail multiple separations (up to 16), which probably brings other psychological processes into play. Mineka et al. (1985) did not find despair following four repeated separations.

After the tradition of Bowlby (1969; 1973), Reite & Capitanio (1985) have viewed attachment as a neurobiologically-based and -mediated behavioral system, requiring species-appropriate experiences for its proper development, and regulating both physiology and behavior of the infant (e.g., Field, 1985; Hofer, 1984, Brazelton, Koslowski, & Main, 1974; Sander, Stechler, Burns, & Julia, 1970). Peer-reared infants provide each other with regulatory support that is qualitatively and quantitatively different from that provided by the mother. Thus, we would expect their "attachment," and hence their response to its disruption, to be fundamentally different.

Boccia & Campos (1987) have suggested that one of the species-appropriate experiences that promotes the formation of an attachment is the emotional security provided by the mother, as a source of reliable information about the environment, and as a model of adaptive responses to ambiguous events which the infant can monitor and imitate. A naive peer would not be able to provide this information, and therefore, a significant psychological component of the foundation for the formation of the attachment bond is missing.

Overall these studies suggest that different social separation paradigms are probably not equivalent in the response elicited or mechanism activated, reflecting differences in both environmental factors and attachment bonds. Separation paradigms might, thus, be systematically manipulated to emphasize one or more of the several underlying phenomena thought to be important in the subsequent response.

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Notes

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